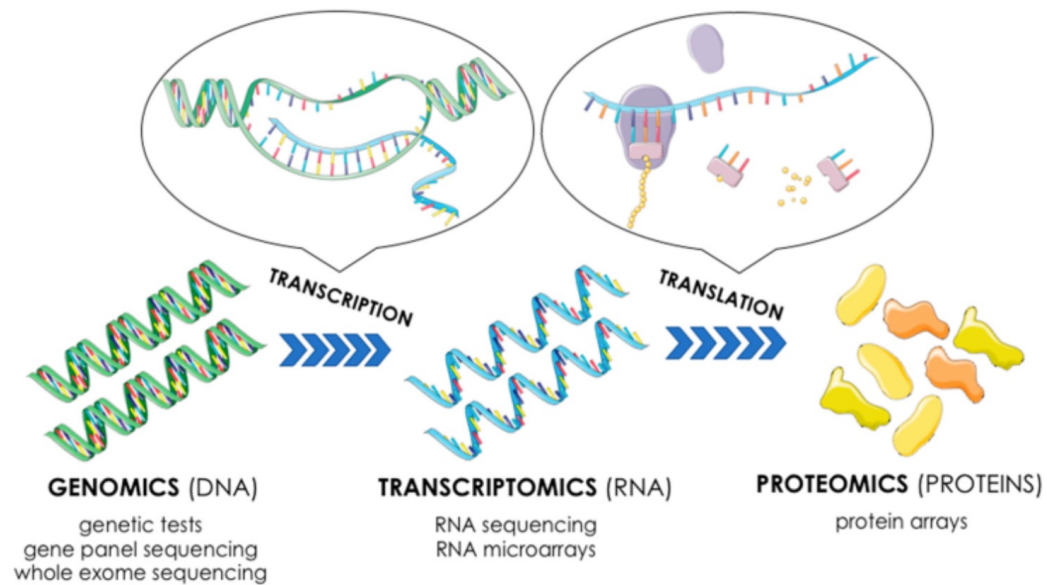


scLong: A Billion-Parameter Foundation Model for Capturing Long-Range Gene Context in Single-Cell Transcriptomics

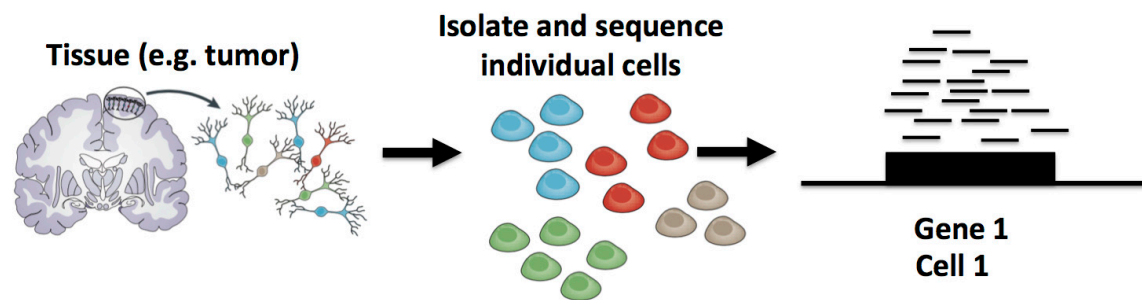
Pengtao Xie
UCSD and MBZUAI

Joint work with Ding Bai, Shentong Mo, and Eric Xing at MBZUAI

Transcriptomics



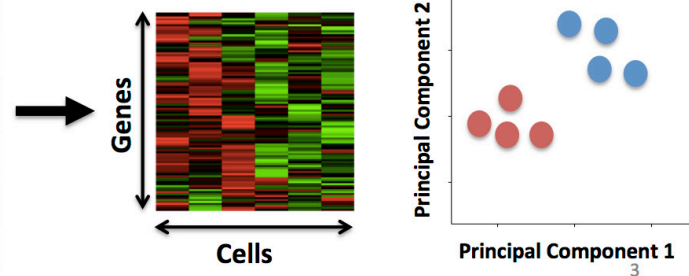
Single-cell RNA sequencing (scRNA-seq)



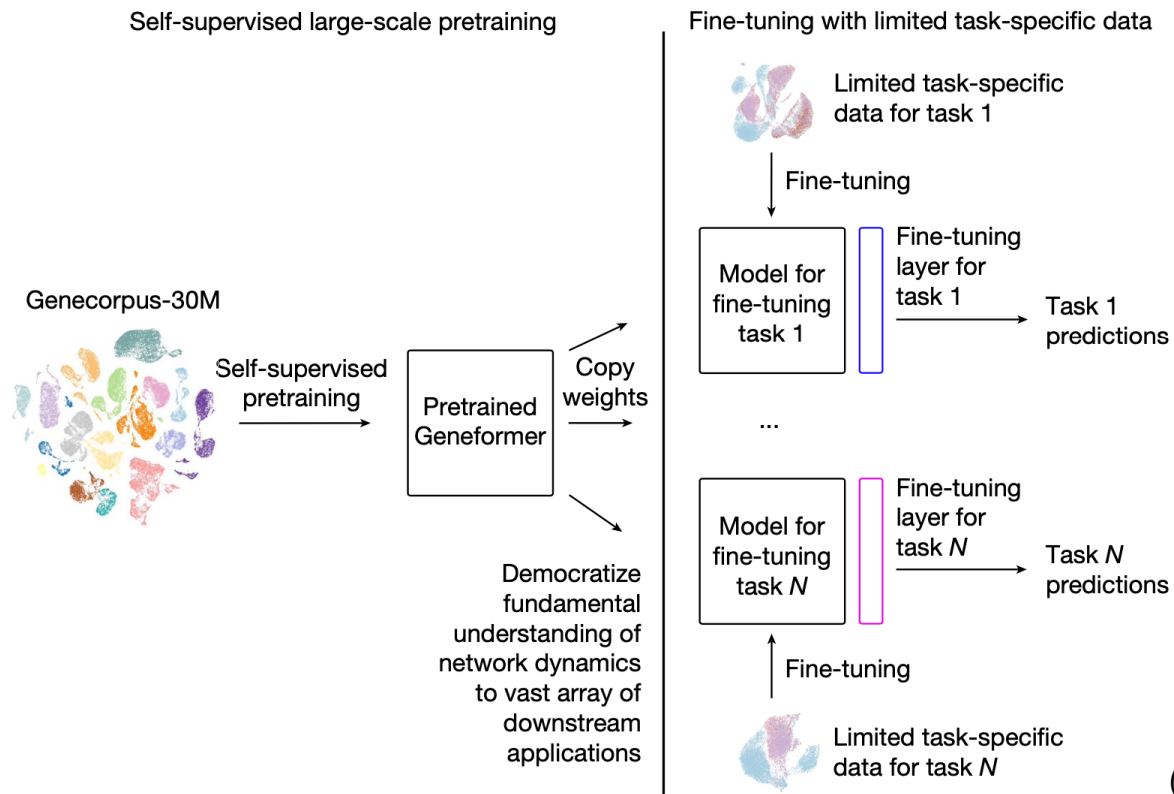
Read Counts

	Cell 1	Cell 2	...
Gene 1	18	0	
Gene 2	1010	506	
Gene 3	0	49	
Gene 4	22	0	
...			

Compare gene expression profiles of single cells



Foundation model

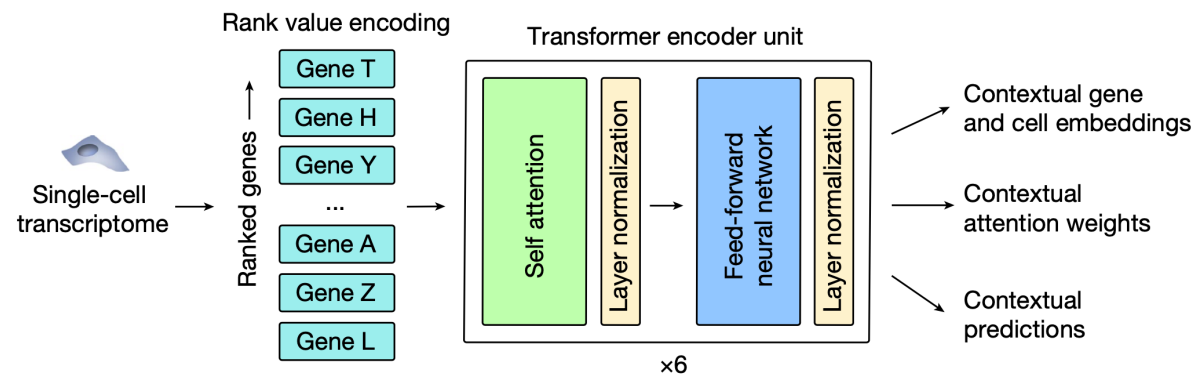


(Theodoris et al.,2023)

Foundation models on scRNA-seq data

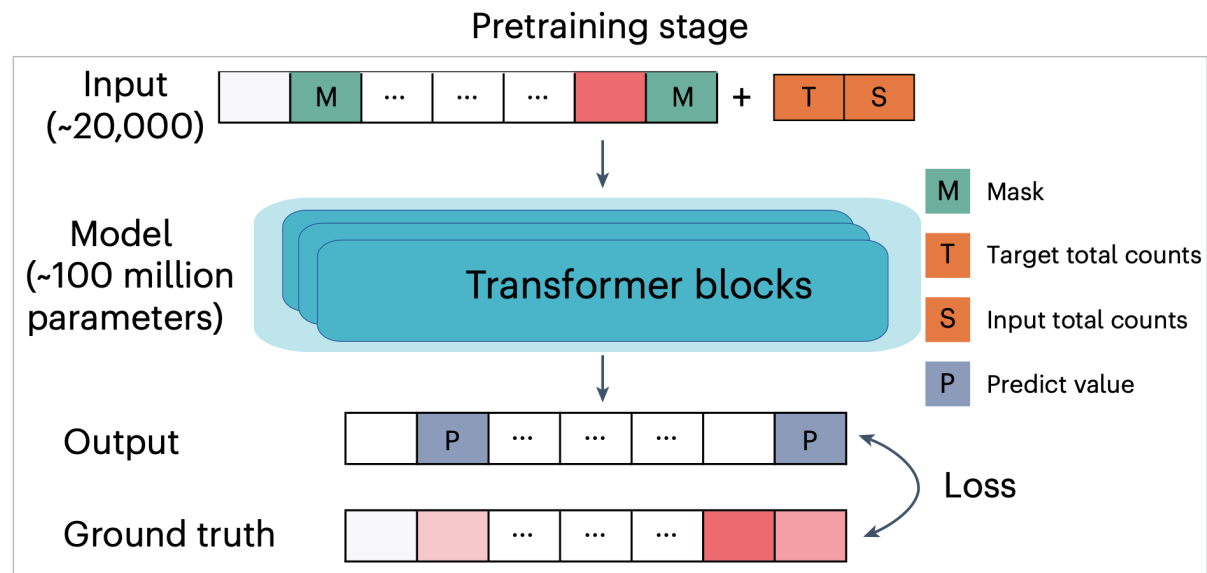
- Geneformer
- scGPT
- scFoundation
- scLong
- ...

Geneformer



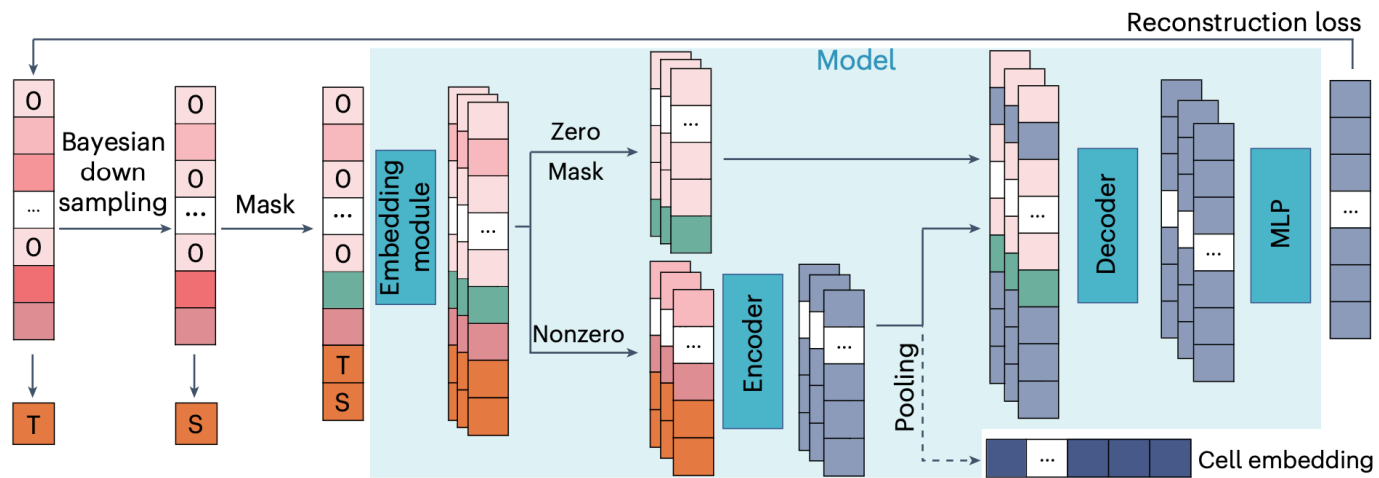
(Theodoris et al., 2023)

scFoundation



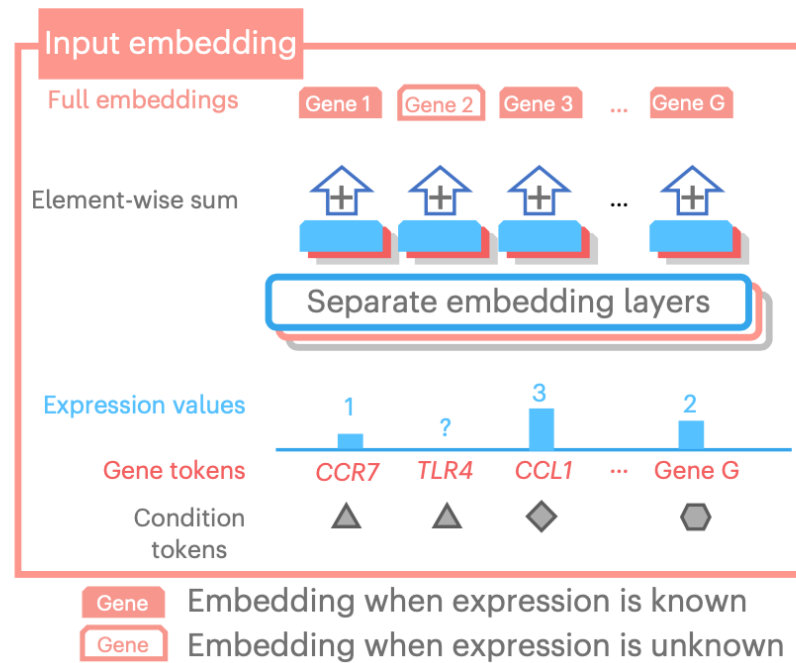
(Hao et al.,2023)

scFoundation



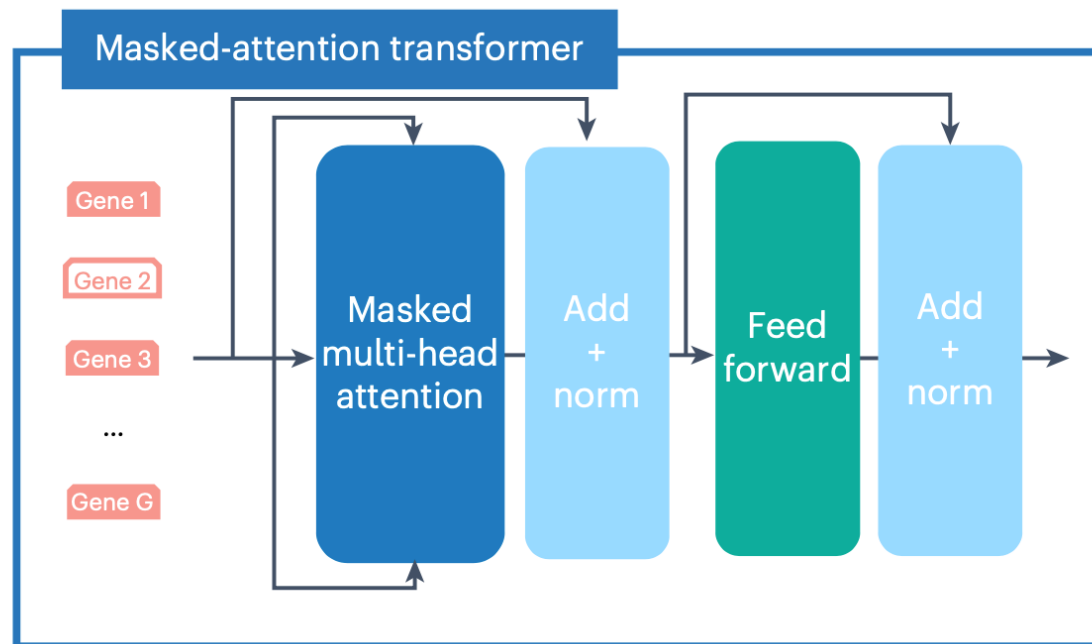
(Hao et al.,2023)

scGPT



(Cui et al., 2023)

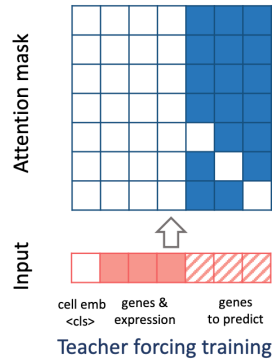
scGPT



(Cui et al.,2023)

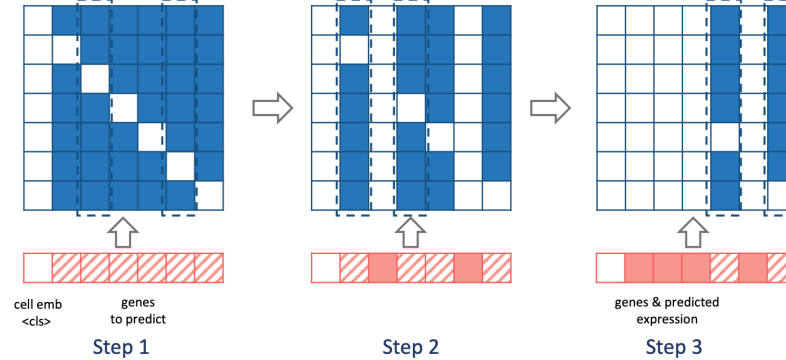
scGPT

A Generative training



Teacher forcing training

B Generation steps during inference



Input embeddings:

- Special <cls> token for cell embedding
- Gene id plus actual or pred expression
- Gene id only

Attention:

- Allowed Positions
- Masked Positions
- The selected positions with high prediction confidence

(Cui et al.,2023)

Limitations of existing foundation models

- Perform self-attention on a small subset of highly expressed genes
 - 2048 in Geneformer and scFoundation, 2000 in scGPT
 - Many lowly expressed genes play essential roles in cellular processes and regulatory networks
 - Miss important regulatory signals and fail to capture long-range gene interactions across the entire genome
- Lack integration of external gene-specific knowledge
 - Rely predominantly on patterns derived solely from gene expression data
 - Fail to capture context related to gene functions and regulatory interactions

scLong: A Billion-Parameter Foundation Model for Capturing Long-Range Gene Context in Single-Cell Transcriptomics

Ding Bai^{1, *}, Shentong Mo^{1, *}, Ruiyi Zhang^{2, *}, Yingtao Luo³, Jiahao Gao⁴, Jeremy Parker Yang⁵, Qiuyang Wu⁶, Digvijay Singh⁷, Hamidreza Rahmani⁸, Tiffany Amariuta^{4, 9}, Danielle Grotjahn⁸, Sheng Zhong⁶, Nathan Lewis^{10, 6}, Wei Wang^{5, 11}, Trey Ideker^{4, 6}, Pengtao Xie^{2, 1, 4}✉, and Eric Xing^{1, 3}✉

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⁸Department of Integrative Structural and Computational Biology, The Scripps Research Institute, La Jolla, CA 92037, USA

⁹Halicioğlu Data Science Institute, University of California San Diego, La Jolla, CA 92093, USA

¹⁰Department of Pediatrics, School of Medicine, University of California San Diego, La Jolla, CA 92093, USA

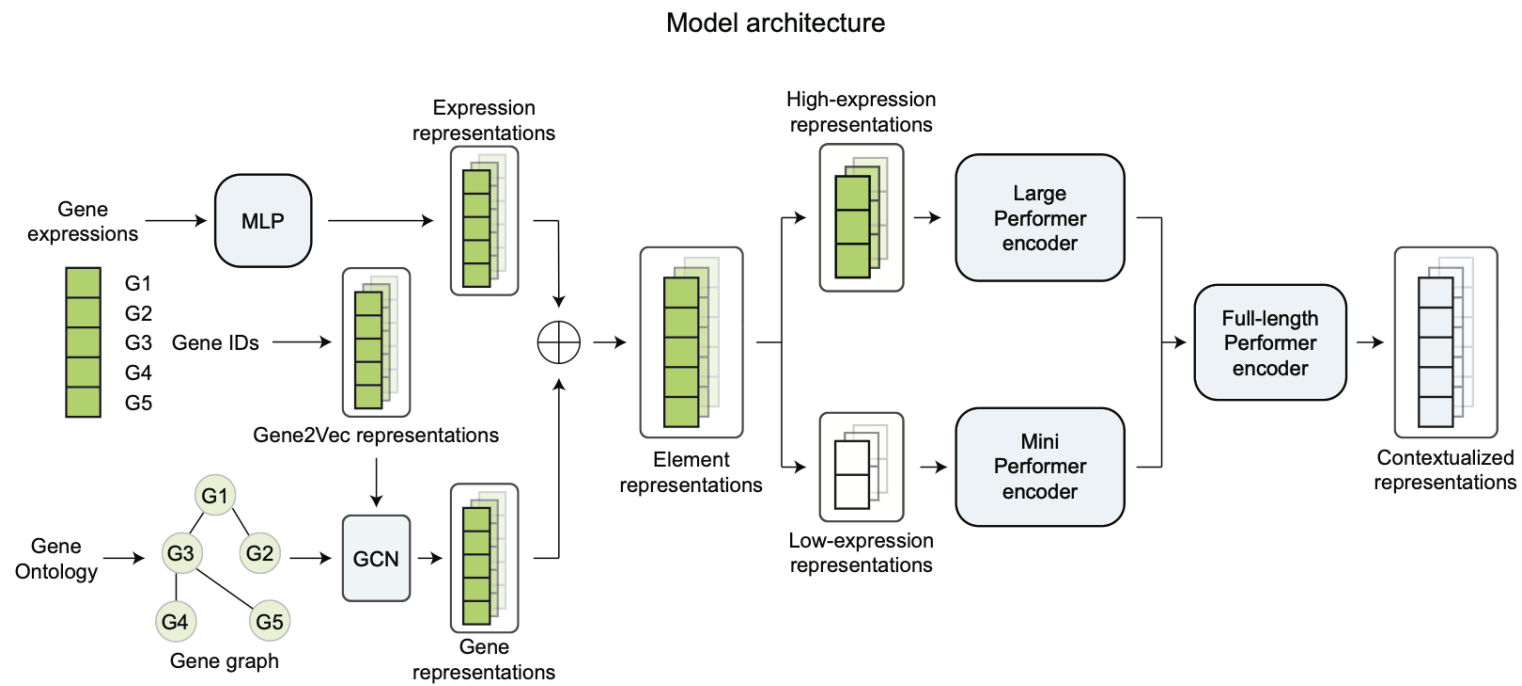
¹¹Department of Cellular and Molecular Medicine, School of Medicine, University of California San Diego, La Jolla, CA 92093, USA

Nature Communications, 2025

scLong

- Perform self-attention across the entire human transcriptome, encompassing around 28,000 genes
 - Capture long-range interactions and dependencies between all genes,
 - Including those with low expression levels that may still play crucial roles in cellular processes
- Integrates external gene knowledge from the Gene Ontology
 - Incorporate hierarchical and functional relationships between genes
 - Provide deeper functional context to its prediction
- One billion parameters
 - Largest to date

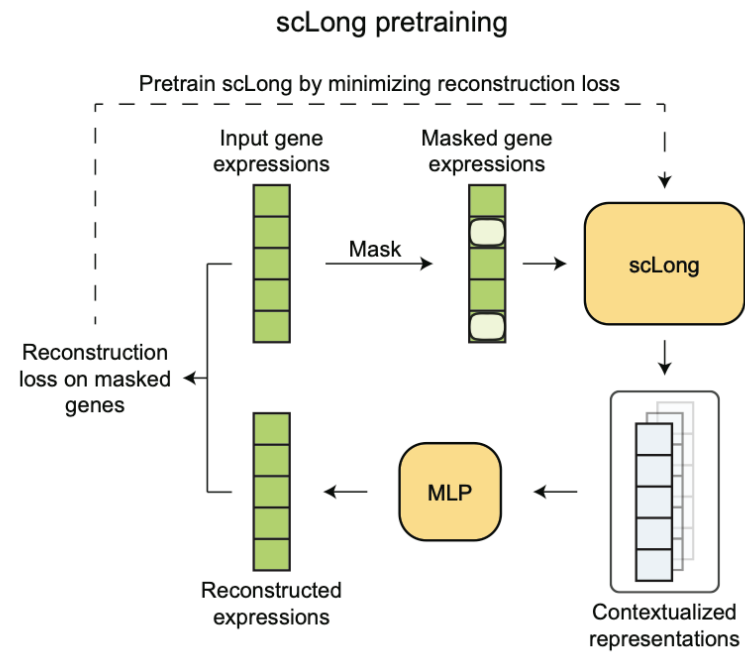
Model architecture



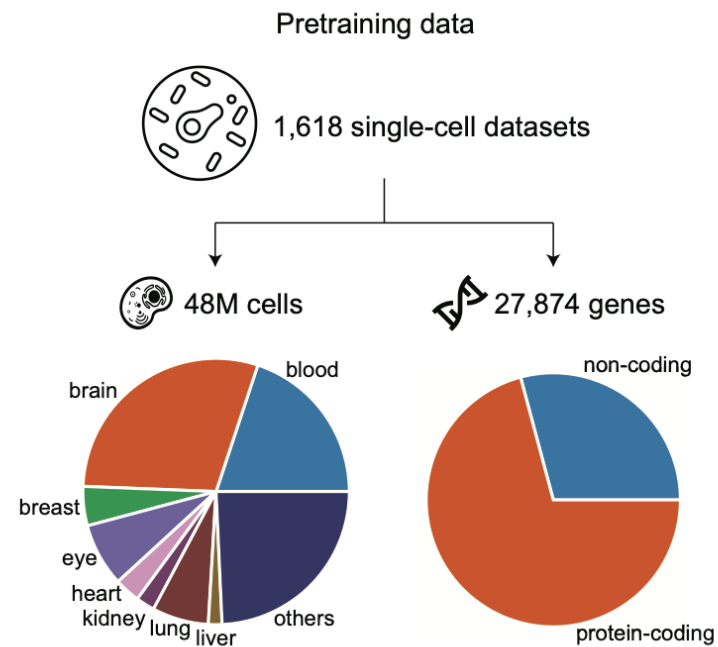
Gene ontology

☐ Gene/product	Gene/product name	Annotation qualifier	GO class (direct)
☐ acp6	acid phosphatase 6, lysophosphatidic		lysobisphosphatidic acid metabolic process
☐ acp6	acid phosphatase 6, lysophosphatidic		lysophosphatidic acid phosphatase activity
☐ acp6	acid phosphatase 6, lysophosphatidic		mitochondrion

Pretraining strategy



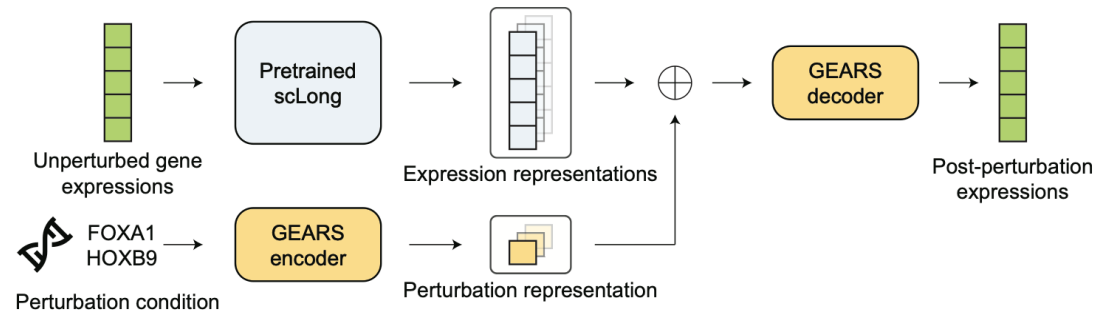
Pretraining data



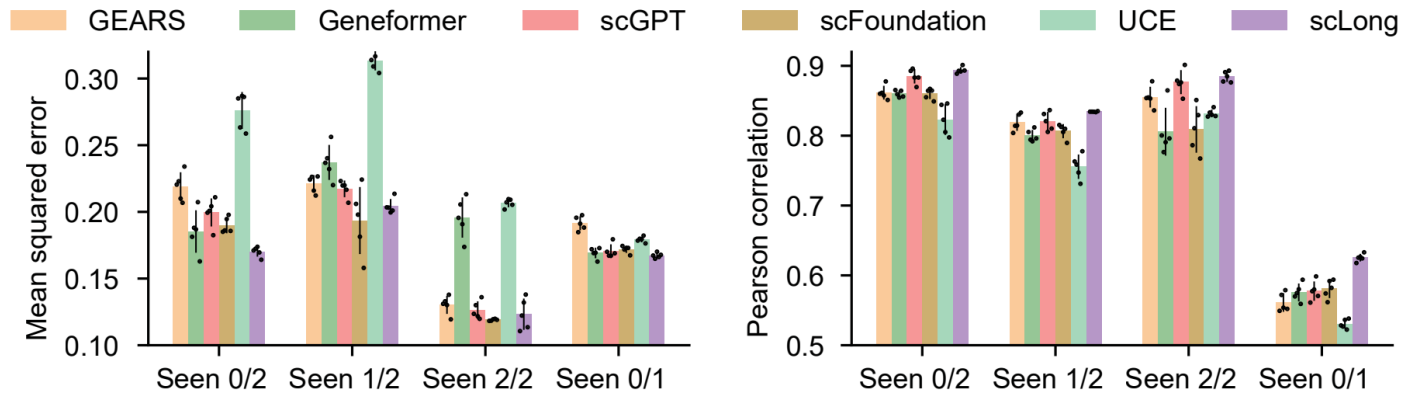
Downstream applications

- Predict transcriptional outcomes of genetic perturbations
- Predict transcriptional outcomes of chemical perturbations
- Predict cancer drug response
- Predict cancer cell line responses to synergistic drug combinations
- Infer gene regulatory network

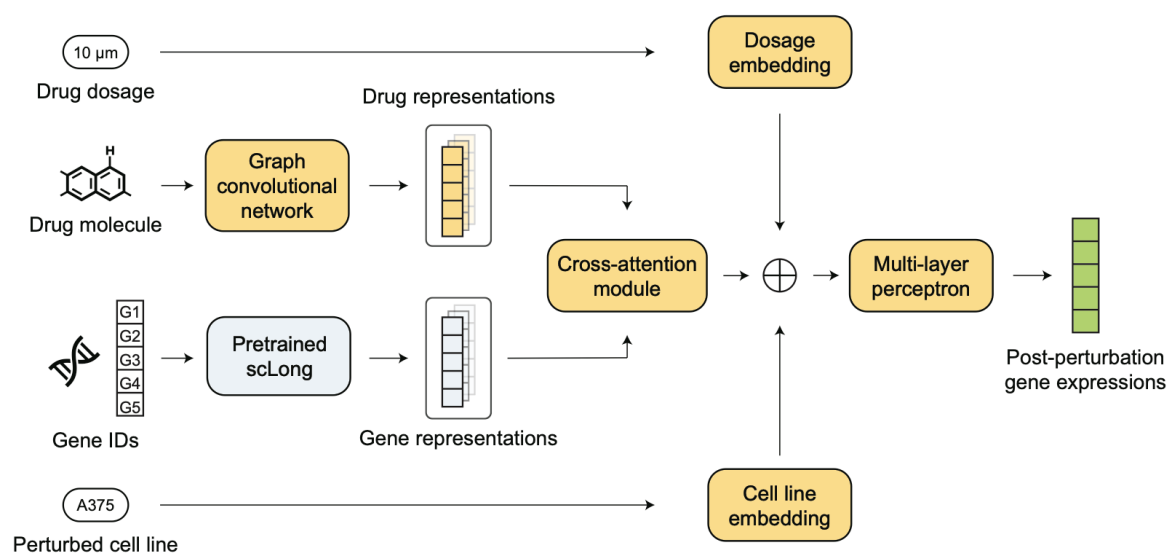
Predict transcriptional outcomes of genetic perturbations



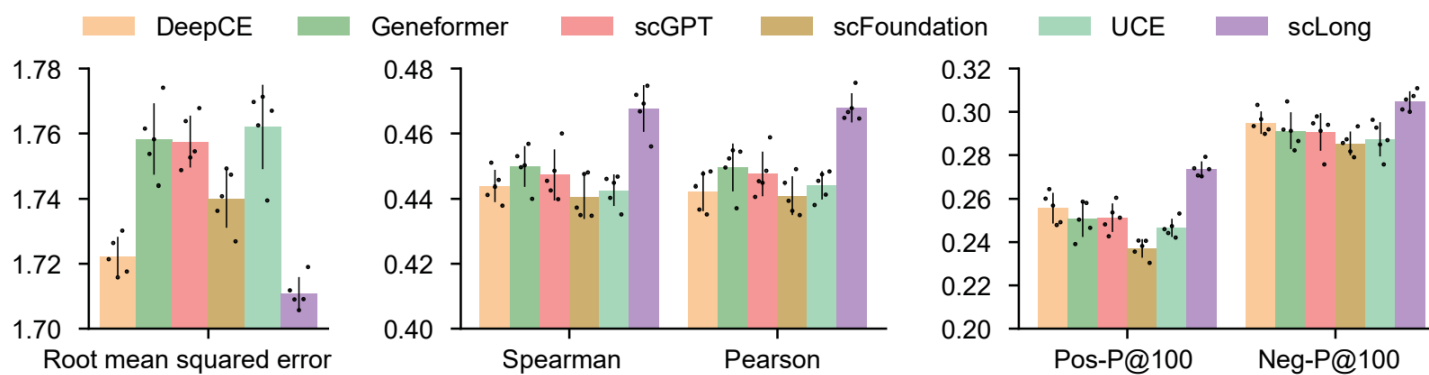
Predict transcriptional outcomes of genetic perturbations



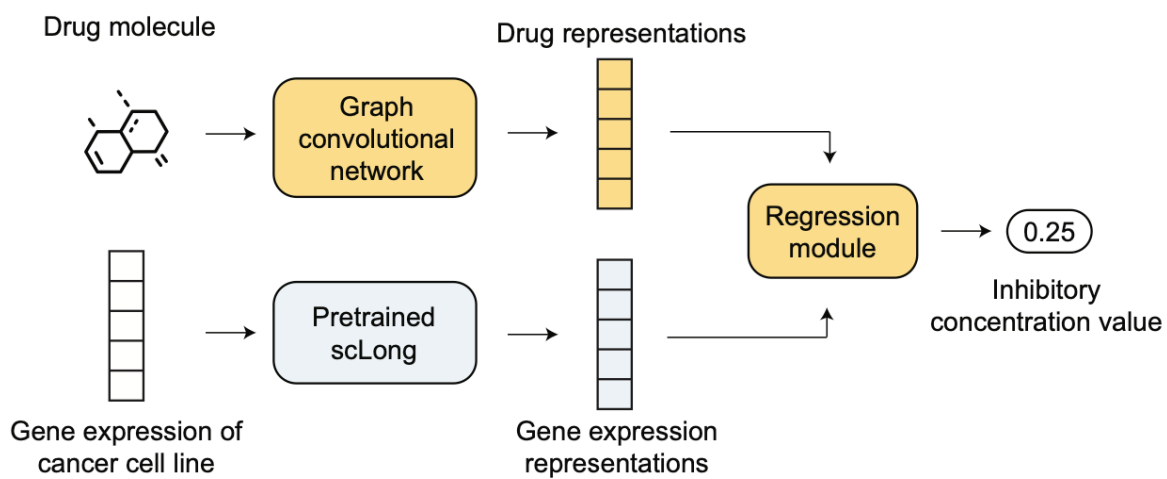
Predict transcriptional outcomes of chemical perturbations



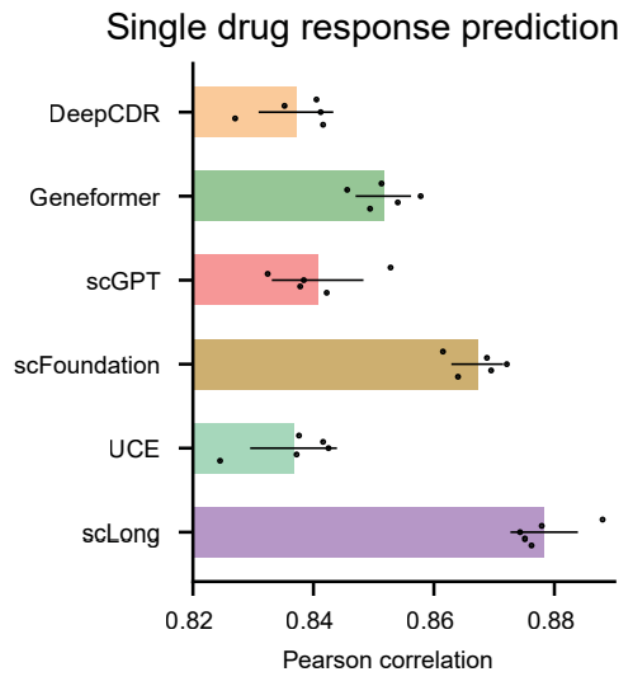
Predict transcriptional outcomes of chemical perturbations



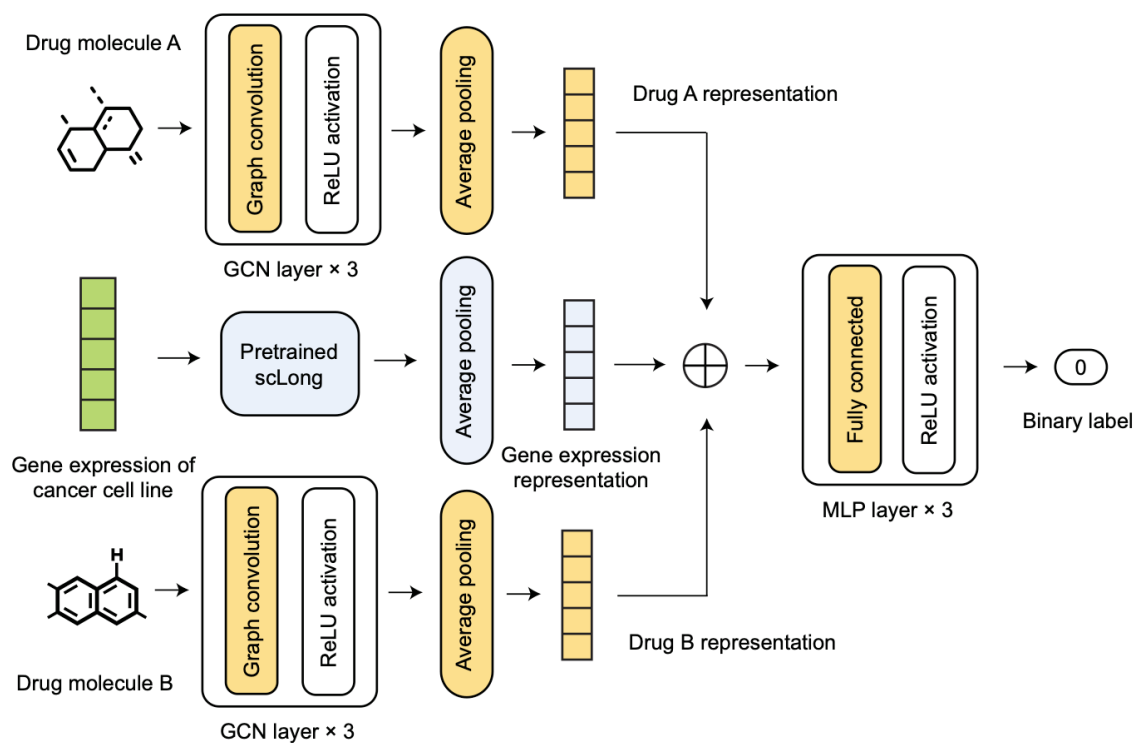
Predict cancer drug response



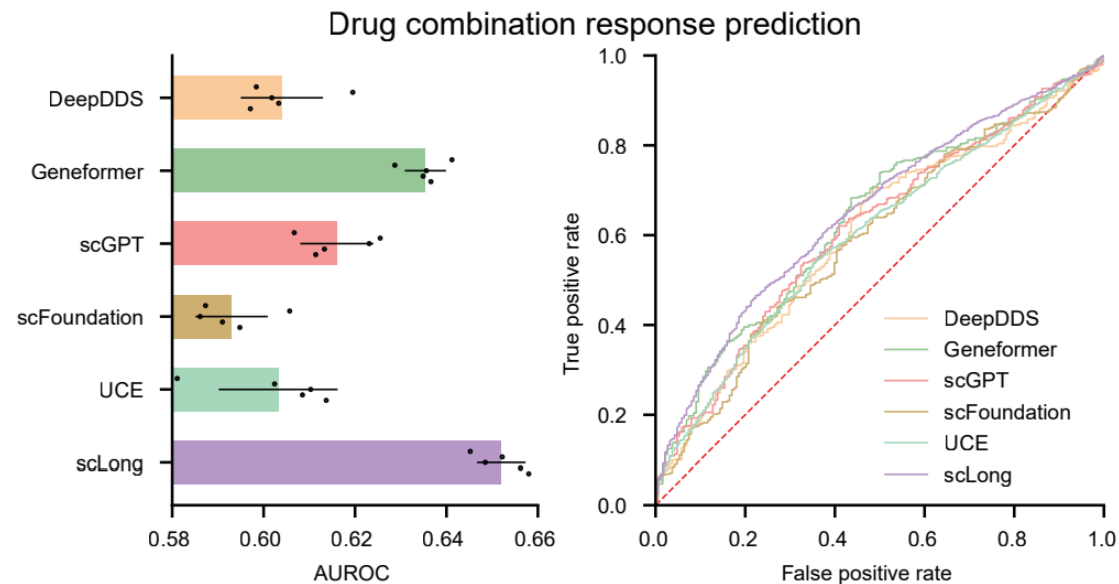
Predict cancer drug response



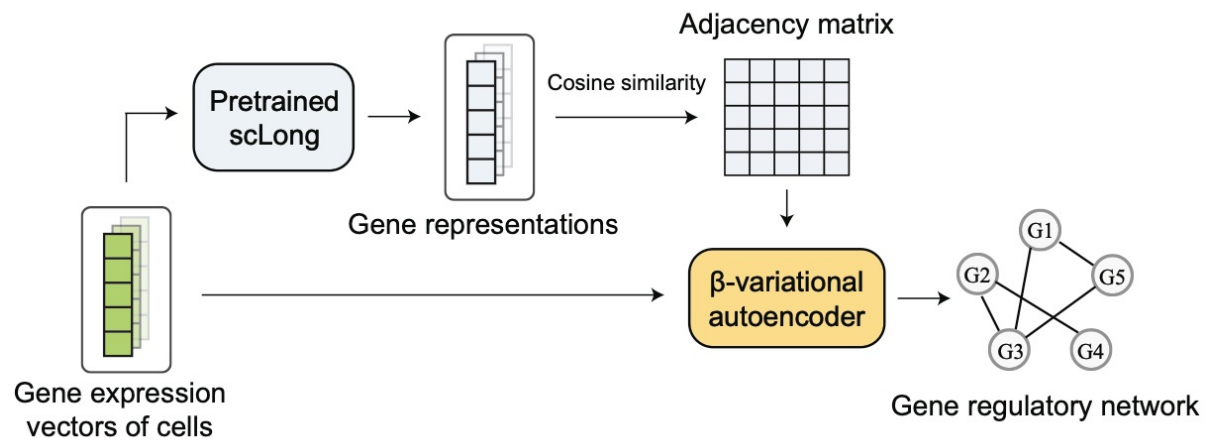
Predict cancer cell line responses to synergistic drug combinations



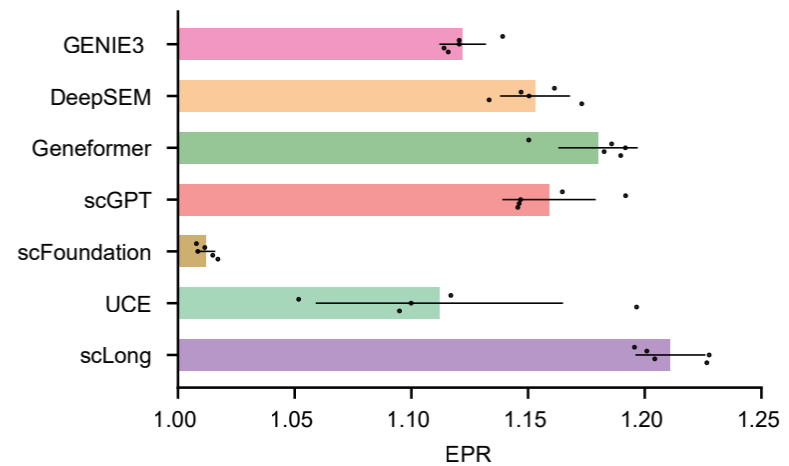
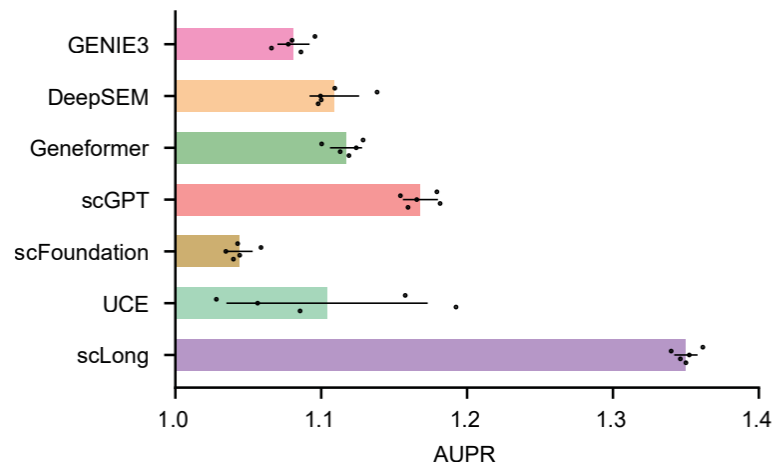
Predict cancer cell line responses to synergistic drug combinations



Infer gene regulatory networks



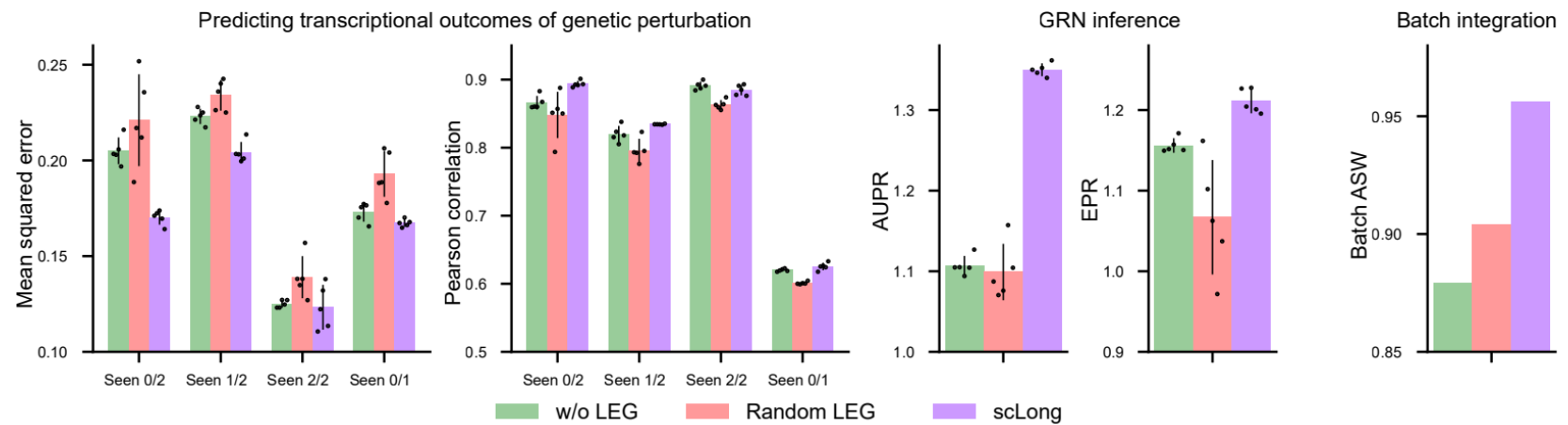
Infer gene regulatory networks



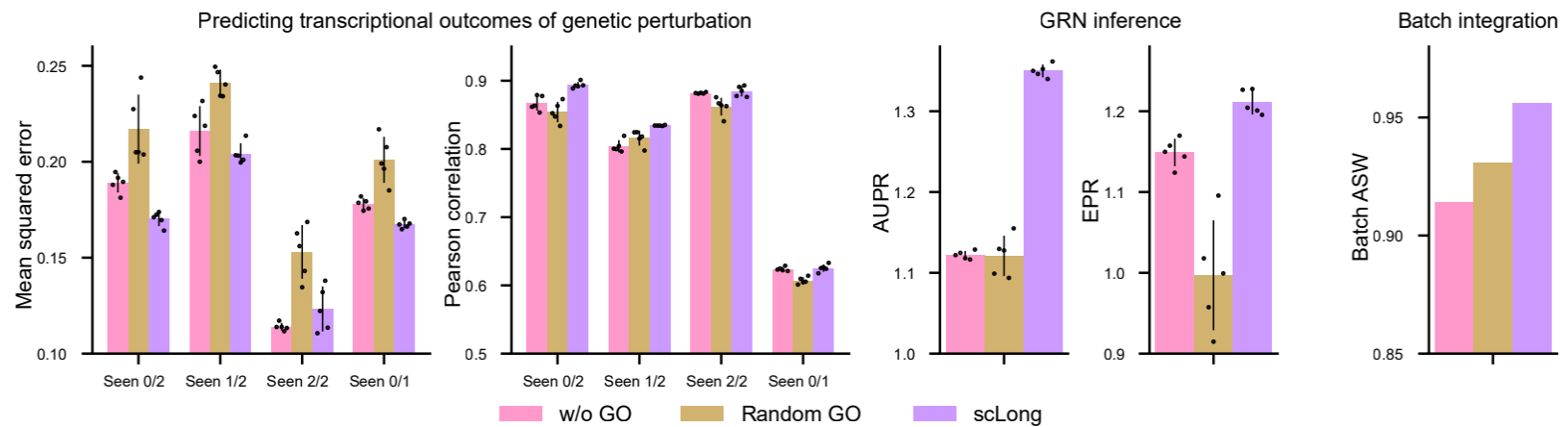
Zero-shot batch integration



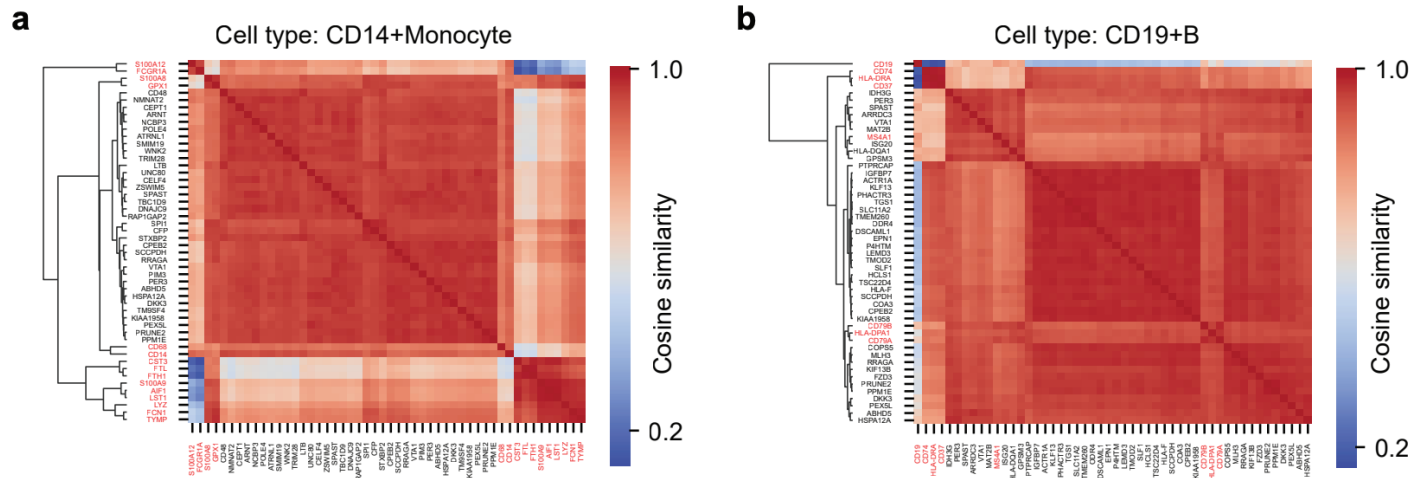
Ablation study: modeling low-expression genes



Ablation study: Integrating Gene Ontology



scLong clusters marker genes associated with different cell types



Conclusions

- scLong
 - Perform self-attention across the entire human transcriptome, encompassing around 28,000 genes
 - Integrates external gene knowledge from the Gene Ontology
 - One billion parameters